HIHO Glove Lab

Build Manual and Shopping List, DIY Finger-Tracking Glove

Based on LucidGloves Prototype 4.1 (open source, MIT license), tracking-only build. Prepared July 6, 2026 for David Bayus. First build: ONE glove (right hand). The second hand costs about a third as much because the tools carry over.

1. What we are building, and why

A fabric glove with five little sensors that measure how much each finger is bent, many times per second. It plugs in over USB (and can go wireless later). In the HIHO pipeline this data will steady the fingers, which are the noisiest part of every mocap take: the cameras see WHERE your hands are very well, but they struggle with exactly how the fingers curl. The glove is the opposite: it knows the curl perfectly and knows nothing about where the hand is. Together they cover each other's weaknesses.

We are building the tracking half of a project called LucidGloves, the most established open-source DIY glove (30,000+ community members, full wiki, video tutorials). The full project adds force-feedback motors so VR players can 'feel' objects. We are skipping all of that: no motors, no power supply. The official docs confirm the glove works fine without them, and they can be added later if the club ever wants haptics.

The learning goal comes first: David builds pair #1 with his own hands to learn soldering and basic electronics, then teaches the build to the 4D ART CLUB. Expect the first glove to be slow and imperfect. That is the point.

2. How it works (the whole trick in one paragraph)

A thin string runs from each fingertip back to a small spool on the back of your hand. The spool sits on a volume-knob component called a potentiometer ('pot'). Bend your finger and the string pulls out, turning the knob. Straighten it and a spring winds the string back in. The spring is recycled from a retractable ID-badge reel, the kind office workers clip to their belts. The knob's position is just a number (0 to 4095) that a small computer board reads 5 fingers at a time and sends up the USB cable. That is the entire sensor: string, spool, knob, spring. Everything else is packaging.

3. Shopping list

Prices are July 2026 ballpark figures from US sellers (Amazon / AliExpress / electronics shops). Check current prices before ordering; the LucidGloves wiki warns they drift. Where packs are larger than needed, that is fine: spares cover mistakes and the club builds later.

Part A: glove parts (builds ONE glove, with spares)

#	Item	Buy	Est. price	What it is
1	B10K rotary potentiometers	1 pack of 10	$8-10	The 'volume knobs' that measure finger curl. You need 5 per glove; a 10-pack covers both hands. Must say 'B10K'.
2	Retractable badge reels	1 pack of 10	$8-12	Cheap office ID-badge reels. We harvest the spring inside. Need 5 per glove; buy 10 for spares + hand two.
3	ESP32 dev board (DevKit style)	1 board (2-packs are common)	$8-16	The small computer that reads the knobs. ESP32 (not Arduino Nano) because it has Bluetooth, so the glove can go wireless later, which matters when performing in the capture volume.
4	Nylon inspection gloves or thin golf/cycling gloves	1 pair	$8-12	The fabric base everything mounts to. Snug fit matters more than brand.
5	Braided fishing line or strong nylon thread	1 small spool	$6-8	The fingertip strings. Any thin, strong, low-stretch line.
6	Dupont jumper wires, female-female, 20cm	1 pack of 40	$6-8	Pre-made wires for first assembly, before committing solder.
7	Hook-and-loop (Velcro) strap + adhesive squares	1 small roll	$6-8	Mounts the printed housing to the glove and board to wrist.
8	Micro-USB or USB-C data cable (match your ESP32)	1	$5-7	Connects glove to laptop. Must be a DATA cable, not charge-only. Check the listing says 'data'.

Part A total: roughly $55-80. (LucidGloves community reports about $30 per hand for tracking-only once tools are owned; the extra here is pack sizes and spares.)

Part B: tools you will own forever (one-time purchase)

#	Item	Buy	Est. price	Notes
9	Temperature-controlled soldering iron kit (e.g. 60W kit with stand, solder, pump, tips)	1 kit	$25-40	Get temperature control, it is the difference between learning and burning. Kits include solder wire and a desoldering pump for undoing mistakes.
10	'Helping hands' holder with magnifier	1	$12-18	Holds the wire so both of your hands are free. Essential, not optional, for a first-time solderer.
11	Learn-to-solder practice kit (any beginner LED kit)	1	$8-12	You solder THIS first, not the glove. Twenty joints of pure practice on something disposable.
12	Basic multimeter	1	$12-15	Answers 'is this wire actually connected?' Claude reads the numbers with you; you just touch the probes.
13	Wire strippers + flush cutters	1 of each (often bundled)	$10-14	Scissors ruin wire; these do not.
14	Heat-shrink tubing assortment	1 box	$6-8	Slides over finished joints so nothing shorts.
15	Safety glasses	1	$5-8	Solder can spit. Non-negotiable.

Part B total: roughly $80-115, once, ever. These same tools cover every future HIHO electronics project (and the club kit list is this exact table).

Part C: 3D printed parts (free, print at home)

The spool housings and finger-cap pieces are free STL files in the LucidGloves GitHub repo (hardware folder, use the '.1' versions, they print easier and last longer). David's Anycubic Kobra Max handles these with room to spare: the parts are small, standard PLA, no supports drama, and the Kobra Max's large bed could fit a full hand's set (or both hands') in a single plate. Still: print ONE finger assembly first as a fit test against a real potentiometer and badge-reel spring before committing a full plate.

Grand total, first glove + all tools: roughly $135-195. Second glove after that: roughly $25-35 (parts only, tools and leftover packs carry over). For scale: Rokoko's gloves are about $2,500 a pair.

4. Learning to solder (the right order)

Soldering is 90 percent patience and 10 percent knowing the order of operations. The rule that makes it work: **heat the parts, not the solder.** Touch the iron to the two metal pieces for a second or two, then feed solder INTO the joint; it flows toward heat like water toward a drain. If you melt solder on the iron tip and dab it on, you get a gray blob that fails in a week (the classic 'cold joint').

Step	What	Done when
1	Set up a ventilated spot (open window or fan), safety glasses on, damp sponge for the tip.	Iron holder is stable and nothing flammable is within arm's reach.
2	Solder the practice LED kit (item 11), every single joint.	Joints look like tiny shiny volcano cones, not dull balls. The kit's lights work.
3	Practice 'tinning' wire ends: strip, twist, touch iron, feed a little solder so the strands become one.	Ten tinned wire ends in a row without frayed strands.
4	Solder wires onto ONE spare potentiometer (there are spares in the 10-pack, mistakes are budgeted).	Multimeter shows the resistance changing smoothly as you turn the knob.
5	Only now, start the glove.	You are calm doing steps 3-4. That is the readiness test.

During bench sessions Claude handles every number: resistor values, which pin goes where, temperatures, measurements. You bring the hands and the patience. Never solder tired.

5. Build overview (the map, not the turn-by-turn)

The turn-by-turn is the official LucidGloves Prototype 4 Building Tutorial on the project wiki, plus LucasVRTech's step-by-step video series on YouTube. Follow those at the bench with this map beside you, and simply SKIP every step involving servo motors, the 5V power supply, or force feedback. That is the entire tracking-only modification.

Stage	What happens	Rough time (first build)
A. Print	Print one finger assembly, test fit, then the rest.	1-2 printer sessions
B. Springs	Open 5 badge reels, harvest the coil springs. (Wear the safety glasses: springs jump.)	1 evening
C. Spool assemblies	Mount each spring + spool + potentiometer into its printed housing. WD-40 on the pots makes them silky.	1-2 evenings
D. Wiring	Connect 5 pots to the ESP32. First with plug-together dupont wires to prove it works, THEN solder the final version. Two wires are shared by all pots; one signal wire per finger.	1-2 evenings
E. Firmware	Load LucidGloves firmware onto the ESP32 from a laptop (no soldering, just software; Claude sits with you).	1 hour

Stage	What happens	Rough time (first build)
F. Glove-up	Velcro the housings to the glove, run strings to fingertip caps, adjust tension.	1 evening
G. Wiggle test	Open the serial monitor, bend fingers, watch five numbers dance. Save a first CSV of you counting on your fingers.	The fun hour

6. HIHO-specific notes (how our build differs from theirs)

Topic	LucidGloves stock	HIHO
Purpose	Live VR hands in SteamVR games	Recorded finger data, merged with camera mocap in Blender
Haptics (motors)	Yes, Prototype 4 default	No. Skip entirely; official docs support this.
SteamVR / OpenGloves driver	Required	Not used. We read the glove's serial data directly and log it per take.
Data path	Streams into the game	Streams up USB; a small logger script saves it as CSV next to the take (club software project, designed later, research-doc-first).
Sync to cameras	n/a	Same visual-marker ritual as the face capture: flash or clap at take start (rig video is silent, so the marker must be visible). Glove CSV rows carry timestamps.
Wireless	Optional	Later upgrade: ESP32 already has Bluetooth; add a small battery, no re-soldering of sensors.

Buying note: nothing here violates the zero-paid-dependencies rule. All software is free and open source (LucidGloves is MIT licensed); the money is hardware only.

7. Links (verify prices and read before ordering)

- LucidGloves repo + STL files: github.com/LucidVR/lucidgloves (hardware folder, '.1' versions)

- Prototype 4 Parts List: github.com/LucidVR/lucidgloves/wiki/Prototype-4-Parts-List

- Prototype 4 Building Tutorial: github.com/LucidVR/lucidgloves/wiki/Prototype-4-Building-Tutorial

- LucasVRTech on YouTube: the Prototype 4 build video series (watch stage-by-stage at the bench)

- Project Discord (30k+ members) linked from the repo: where beginners get unstuck.

Skipped on purpose: Prototype 5 (still BETA, no stable hardware files, wrong choice for a first solder project). Re-check its status when the club build happens.